

ROM-fed Fault-Injection Shell — Vivado Elaborated Schematics

Local demonstration / GitHub schematic companion | Vivado 2022.1 | part `xc7vx690tffg1761-2`

Top: `top_fi_rom_*` (example: `top_fi_rom_s3` → DUT `top_s3_subfft_ecc`)

1. Purpose

These figures are Vivado **Elaborated Design** schematics of the **ROM-fed FI shell** used for local board-style demonstration. On-chip stimulus ROM provides FFT input beats; UART is used **only** for fault-injection (ARM/FIRE) commands. This shell is **not** a paper fair-kernel resource top (`vivado/K_*` tables remain authoritative). Do not treat pin counts or utilization from this shell as manuscript numbers.

2. Architecture reading of the two schematics

Figure 1 (shell top) shows the intended separation of paths:

- **Control / FI path:** `uart_rx` → `u_uart` (`uart_rx_simple`) → `u_dec` (`uart_cmd_decoder`) → `u_fi` (`fi_controller`) → DUT FI ports (`fi_fire_pulse`, `fi_arm_*`, `fi_site_mem`).
- **Data / stimulus path:** `u_rom` (`stim_rom`, init from `.mem/.coe`) ← addressed by `u_feed` (`stim_feeder`) → DUT parallel inputs (`in_valid`, `in_last`, lane `in*_re/im`).
- **External ports** stay small: `clk`, `rst`, `uart_rx`, plus `status` / `out_valid` / `out_last`. Wide FFT I/O remains *inside* the chip.

Figure 2 (DUT interior) is the opened hierarchy under `u_dut`: a left-to-right chain of per-stage blocks (ten radix-2 SDF stages for the 1024-point kernel). Clock/reset fan out to each stage; data and valid handshake propagate stage by stage. This view is the structural counterpart of the per-stage recovery narrative; FI hooks target declared stage/site boundaries rather than arbitrary I/O pins.

Note: Screenshot resolution follows the Vivado schematic canvas. For repository packaging, place this PDF under the `public schematic/` folder alongside other architecture figures.

Figure 1 — ROM-fed FI shell (elaborated top)

Instances: u_uart, u_dec, u_fi, u_rom, u_feed, u_dut.

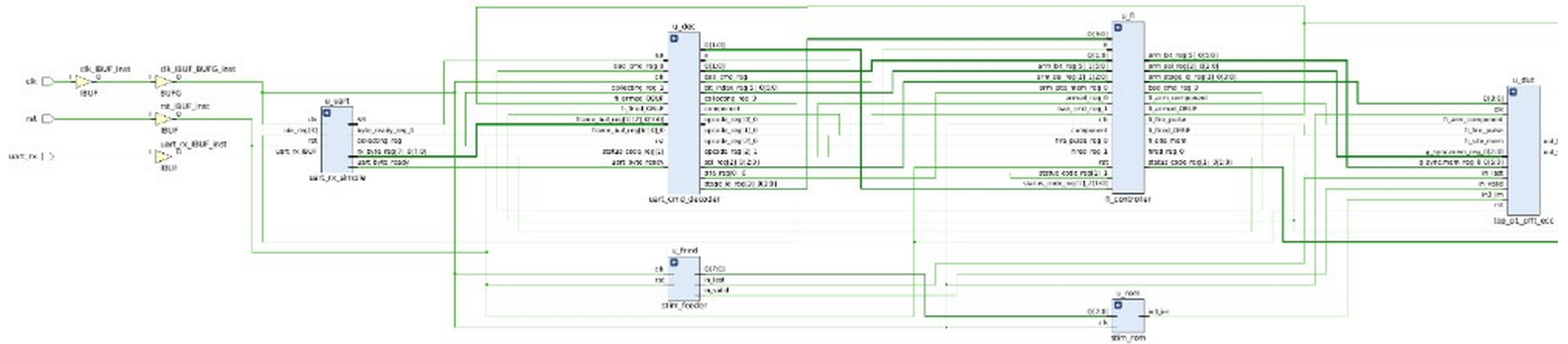


Fig. 1. Vivado elaborated schematic of the ROM-fed FI shell. UART path is FI-only; stimulus ROM feeds the DUT data path.

Figure 2 — DUT stage pipeline (hierarchy under u_dut)

Per-stage SDF chain inside the kernel top (example DUT: top_s3_subfft_ecc).

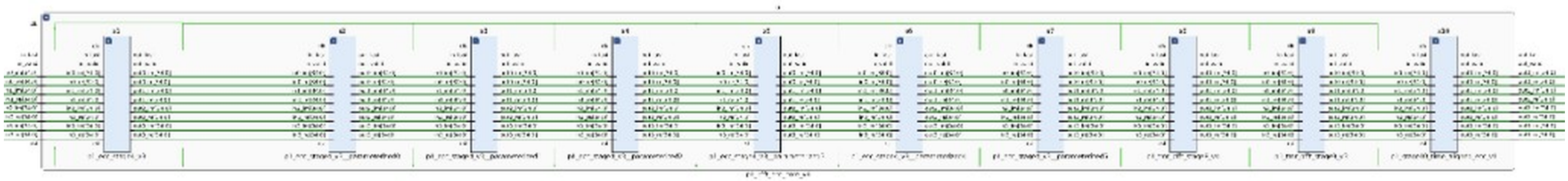


Fig. 2. Vivado schematic of the DUT internal stage pipeline (stages cascaded left → right).

Provenance. Captured from Vivado 2022.1 Elaborated Design views of fi_rom_top_v1/ROM_* projects. Stimulus init files: fi_func_sim_v1/common/coe|mem/*_input.*. Manuscript fair-kernel RTL remains under artifact_v1/*_src/vivado/K*.